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Understanding Readability of Large Language Models Output: An Empirical Analysis

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Abstract

Recently, Large Language Models (LLMs) have seen some impressive leaps, achieving the ability to accomplish several tasks, from text completion to powerful chatbots. The great variety of available LLMs and the fast pace of technological innovations in this field, is making LLM assessment a hard task to accomplish: understanding not only *what* such a kind of systems generate but also which is the *quality* of their results is of a paramount importance. Generally, the quality of a synthetically generated object could refer to the reliability of the content, to the lexical variety or coherence of the text. Regarding the quality of text generation, an aspect that up to now has not been adequately discussed is concerning the *readability* of textual artefacts. This work focuses on the latter aspect, proposing a set of experiments aiming to better understanding and evaluating the degree of readability of texts automatically generated by an LLM. The analysis is performed through an empirical study based on: considering a subset of five pre-trained LLMs; considering a pool of English text generation tasks, with increasing difficulty, assigned to each of the models; and, computing a set of the most popular readability indexes available from the computational linguistics literature. Readability indexes will be computed for each model to provide a first perspective of the readability of textual contents artificially generated can vary among different models and under different requirements of the users. The results obtained by evaluating and comparing different models provide interesting insights, especially into the responsible use of these tools by both beginners and not overly experienced practitioners.

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Keywords: LLMs; Text Generation; Readability; Text Quality Indexes; Text Evaluation Metrics; Generative AI; Artificial vs Hand-crafted generation.

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1. Introduction

Recent advancements in Large Language Models (LLMs), have thrust generative Artificial Intelligence (AI) models into the spotlight for a wider audience. While Generative Adversarial Networks (GANs) [1] and Variational Autoencoders (VAEs)[2] have existed for some time within the research community, LLMs have provided compelling evidence of their potential to a broader user base. Among the large variety of objects that LLMs can provide by their generation mechanisms, Natural Language Processing (NLP) is almost certainly the most impacted field, e.g., ranging from automated content generation and machine translation to an empowered information retrieval and to sentiment analysis. These models, trained on vast amounts of text data, possess remarkable capabilities in producing human-like text content. However, amid the excitement surrounding their capabilities, a critical concern emerges: how to evaluate and assess the quality of artificial objects automatically, without the human expertise in the loop.

More generally, evaluating the quality of textual contents is an old problem rooted in linguistic theory; a subset of the most adopted dimensions for the quality analysis of textual objects are: (1) **grammar and mechanics** (including both grammatical errors checks and readability scores), (2) **vocabulary and style** (covering lexical diversity, sentence length variation, etc.), and (3) **coherence and cohesion** (dealing with co-reference resolution and logical flow analysis). Hence, evaluating automatically generated contents can offer not trivial challenges too. So far, the most attention of the research has been addressed on *what* is possible to generate by such kinds of systems and to improve the results of the process generation. However, the concept itself of "improvement" and how it could be evaluated has not been clearly stated yet in a systematical framework: in fact, very little prominence has been paid to this latter aspect. First, it should be clarified: (1) what we mean for the quality of an artificially generated text, (2) how can we measure it (ordinal and/or quantitative scales), and (3) what kind of measures we should perform to estimate it. Providing the most suitable quality framework is affected by the kind of artifact: some of the affecting factors are — but not limited to — reliability of the content (both for the textual and visual contents [3, 4]), the readability, the lexical variety, and the text coherence.

Indeed, while LLMs excels in generating coherent and contextually relevant texts, they often fall short in producing truly readable and understandable contents. This discrepancy between the fluency of generated text and its actual readability poses significant challenges across multiple domains, including content creation, education, journalism, and user interfaces. Anyway, the *readability* of textual artefacts represents an aspect that up to now has not been adequately discussed. Analysing and understanding the factors contributing to the readability issues of LLM-generated text becomes imperative. Factors such as syntactic complexity, semantic coherence, and stylistic inconsistency play crucial roles in determining the readability of generated content. Moreover, the lack of control over the tone, style, and vocabulary of generated text further exacerbates the readability challenges, especially in applications where clarity and coherence are paramount.

It should also not be undervalued that the readability problems associated with LLM-generated texts have far-reaching implications. In educational settings, poorly readable content may hinder comprehension and learning outcomes, impacting students' academic performance. In journalistic and content creation domains, readability directly influences audience engagement and retention. Additionally, in user interface design and chatbot interactions, readability determines the effectiveness of communication and user experience. Addressing the problems of readability in LLM-generated text requires a multidisciplinary approach combining insights from linguistics, cognitive science, and computer science. Developing techniques to enhance readability while preserving the fluency and coherence of generated text is essential for realising the full potential of LLMs in diverse applications.

In this work, we delve into the intricacies of readability challenges associated with LLM-generated text. We explore the underlying factors contributing to these challenges and examine existing approaches aimed at improving readability. Through empirical analysis and case studies, we aim to provide insights useful to understand mechanisms to implement in LLMs for mitigating readability issues in LLM-generated content, thereby advancing the usability and effectiveness of these powerful language models in real-world applications.

The paper is structured as follows: Section 2 reviews the main papers of the scientific literature and also recalls useful background information; Section 3 describes the methodology followed in the work; Section 4 describes the setting and the different experiments run in this work, along with a preliminary discussion about the results of the experimental campaign, extracting some lesson learned; Section 5 ends the paper and sketches some future research directions.

2. Background and Related Work

At the best of our knowledge, there are currently very few research works relate to the objective of the present paper. The study described in [5] evaluated the impact of the contents generated by LLMs in the field of education to explore the potential of a precise system, namely the OpenAI Chat GPT3.5[6] to aid student learning. It explores the opportunity to tailor the LLMs to produce high-quality contents, where the quality is defined as highly personalized reading comprehension exercises for middle school English learners in China. In [7], a study that explores a readability assessment and a text simplification for generating science documents targeting a multilingual educational context was proposed. Different types of characterisation methods related to text classification and simplification, following a set of pedagogical approaches, were evaluated. Even if it is precisely focused on three languages (Basque, Spanish and English), it suggests an extendable methodology for teachers to follow so that they can adjust and assess materials according to the student profiles in the classroom. The work of [8] represents the closest paper to our study: it focuses on approaches for generating educational materials with different levels of readability by using LLMs. It assessed the capability of some LLMs, to generate content at various readability levels through zero-shot and few-shot prompting. Considered LLMs are: GPT-3.5[6], LLaMA-2 70B[9], and Mixtral 8x7B[10]. Different levels of readability have been obtained by adopting text simplification and transformation rules; while increasing the readability of a document by the means of a smaller set of words and a simplified vocabulary, the context preservation is a fundamental aspect to consider. This latter aspect is among the insights of the study proposed in this work, where we aimed to investigate according to which criteria different LLMs would take a decision to cut contents and simplify complex linguistic structures into other simpler ones.

To this aim, several well assessed text quality evaluation indexes are available from the literature: a subset of them can be adopted to assess various aspects of text, from readability and fluency to coherence and informativeness.

With the advent and advances in AI, automatic text generation tools, such as the ones based on the LLM, have provided another source of artificial but realistic content. Among these, the textual documents can space from narrative texts to argumentative ones, including descriptive and creative ones, such as poems and songs, short stories and fables, technical reports and dissertations.

Naturally, the problem of determining the degree to which the artificial text is "correct" is a challenge that cannot be solved in a bit. First, as it also happens when a human-crafted text is evaluated, the meaning of "correctness" for a text should be clearly stated: this feature can refer to the syntactical correctness of the text, or it may refer to the content of the text that is matching the topic that was requested, or both. Beyond the correctness, "completeness" and the "lexical richness" are other two typical features for stating the overall quality of a manuscript. Evaluating the readability of text generated by language models is a complex problem, implying a trade-off between human (subjective) judgments and automated (objective) metrics.

Dealing with the problem requires a multifaceted approach able to account for the various dimensions of readability; hence, it can be summarised in:

- *Lack of Consensus on Evaluation Methods*: there is no one-size-fits-all approach to assess how good the model's text is. It is, usually, compared to a human-written example, but this benchmark can be subjective.
- *Metrics do not Capture Readability Nuances*: common metrics, like perplexity, measure how likely a sentence is from the model's training data, not how easy it is for a human to understand.
- *Subjectivity*: readability is inherently subjective and varies based on the reader's background, language proficiency, and familiarity with the topic. What one reader finds easy to read, another might find difficult
- *Existing Metrics*: traditional metrics, such as the Flesch-Kincaid readability test, are often based on surface-level text features like sentence length and word complexity. These metrics may not fully capture the nuances of readability in the context of automatically generated text, including coherence, logical flow, and contextual relevance.
- *Context and Cohesion*: automatically generated text can sometimes be syntactically correct but semantically incoherent. Evaluating readability requires assessing whether the text makes sense in a broader context, which is difficult to quantify with existing automated tools.

- *Human Evaluation*: while human evaluations are the gold standard for assessing readability, they are time-consuming, expensive, and not scalable for large volumes of text. Balancing the need for human insight with the efficiency of automated methods is a significant challenge.
- *Complexity of Language Models*: advanced language models can produce text that mimics various writing styles and complexities. This versatility makes it difficult to apply a one-size-fits-all readability assessment, as different texts might require different evaluation criteria.
- *Adaptation of Metrics*: there is a need to adapt existing readability metrics or develop new ones that can better handle the specific characteristics of machine-generated text. This includes accounting for the potential for subtle errors and the overall coherence of longer passages.

2.1. Readability Indexes

Readability is the ease with which a reader can understand a written text. In natural language, the readability of text depends on its content (the complexity of its vocabulary and syntax). It focuses on the chosen words, and their usage in sentences and paragraphs.

To engage the readers, it is critical to present information to them that they will gladly keep reading and be able to understand clearly. Therefore, it is necessary that the content be easy enough to read and understand, as readable as possible. Anyway, providing a system to measure and assess the "readability" of textual contents is not trivial: there are various available difficulty scales with their difficulty determining formulae.

The literature provides several well-reviewed and recognised readability indexes that stand as critical tools for assessing the ease of reading and comprehension of a text [11]. The study described in [12] highlights how the lack of readability can be intentional to obfuscate information and drive misleading information. The problem of automatic detection of fake news both generated by machines or humans provides a notable example of textual contents, which exhibit a low level of readability, intending to confuse the readers [4, 13]. The exploitation of intentionally generating textual contents with low levels of readability can be considered a malicious action belonging to the soft security attacks. Means to perform this effect span from LLM poisoning to driving and biasing the text generation process to obtain specific results. Some works dealing with this specific aspect are [14, 15].

Focusing back on the various readability indexes available from the linguistic literature, the Flesch Reading Ease (FRE) Index is one of the most widely used[16]. Other very common readability indexes are represented by the Gunning Fog (GF) Index [17], the Coleman-Liau Index [18], and the Automated Readability Index [19].

In NLP, sometimes it is required to analyse words and sentences to determine the difficulty of the text. Readability Scores are generally grade levels on particular scales, which rate the text as to what the difficulty of that particular text is. It assists the writer in improving the text to make it understandable for a larger audience, thus making content engaging. In short, current methods might not fully capture the readability aspect of AI-generated text.

From the literature, various traditional readability formulae are available for readability score evaluation. Although it is possible to identify the advantages of such formulas, it is necessary to note some disadvantages of their use. On one hand, readability formulas offer several advantages. They measure the grade-level required to read a given text, providing writers with essential information to effectively reach their target audience. By knowing in advance whether the target audience can understand the content, writers can tailor their material accordingly. Further, these formulas are easy to use, and creating readable text helps attract a larger audience. On the other hand, the existence of many different formulas can lead to wide variations in results for the same text. The application of mathematical models does not always capture the essence of the literature, so is not always ideal. Additionally, these formulas cannot measure the complexity of a word or phrase precisely, making it difficult to identify specific areas needing correction. Among the most adopted readability scores, in this study a subset made of three indexes has been considered, such as:

- *Dale-Chall Readability Formula*: this is a widely used method for assessing the reading difficulty of a text. It primarily focuses on vocabulary and sentence length, not considering other factors like sentence structure or grammar. Indeed, it analyses the vocabulary used in a text dividing the words into two types, the "difficult" ones are not contained in a list of $3,000$ words considered familiar to fourth-grade students in the US. The interpretation of the score assigned to each word can be explained by considering that if the score is higher, the

text is considered more difficult; therefore, readability is lower. Several online tools and calculators are readily available to easily calculate the Dale-Chall score of a text [20].

- *The Gunning Fog Formula*: given an English text, the GF formula calculates the number of years of formal education required for a person to understand the text on first reading easily [26]. In particular, this formula considers text length (total words= ts and total sentences = ts) and word complexity as a characteristic related to the number of syllables (complex words= cw), as is the case for the English language. Since this is not the same for all languages, the authors in [27] show how it is possible to modify the GF equation, given by $GF = 0.4[(tw/ts) + 100(cw/tw)]$, by adapting it to other languages, in their case to Spanish.
- *Flesch's Scores*: two widely used readability formulas were developed by Rudolf Flesch: FRE, and the Flesch Kincaid Grade Level (FKGL). They are designed to evaluate the readability of English texts by measuring sentence length (total words= ts and total sentences = ts) and word complexity (total syllables= tsl) but with two different purposes. In particular, FRE, expressed by $FRE = 206835 - 1015(tw/ts) - 84.6(tsl/tw)$ is used to determine how easy a text is to read. On the other hand, the FKGL, determined by $FKGL = 0.39(tw/ts) + 11.8(tsl/tw) - 15.59$ is used to translate the FRE score into a US school grade level "measure" used for the Dale-Chall formula). According to the literature, those two indexes are the most preferred readability formula [30] even though it may present some limitations [31].

3. Methods and Materials

To investigate the problem of the readability of automatically generated texts by LLMs, we employed a systematic methodology aimed at evaluating the performance of various LLMs across different readability metrics. The methodology comprised three main steps: (1) selection of LLMs, (2) development of an evaluation framework, and (3) design of experimental tasks.

1. Selection of LLMs: We initially identified a pool of open-source LLMs available in the research community. From this pool, we selected several LLMs that represented a range of architectures, training datasets, and model sizes. In detail, the considered LLMs are: Microsoft Phy[21], Mistral[22], OpenAI Chat-GPT3.5[6], and Meta Llama3 with two different sets of hyperparameters [23, 24]. The selected LLMs encompassed both state-of-the-art models and baseline architectures to provide a comprehensive assessment of readability performance across different model complexities.

2. Development of an Evaluation Framework: We constructed an evaluation framework based on well-established readability indexes from computational linguistics literature. Central to this framework was the inclusion of readability metrics such as the Flesch Indexes (both FRE and FKGL), which measure the complexity of written text based on factors like sentence length and word syllables. Additionally, we incorporated other widely used readability metrics to capture various aspects of text complexity, including GF and the Dale-Chall formulas.

3. Design of Experimental Tasks: With the selected LLMs and evaluation framework in place, we designed a series of experimental tasks to assess the readability of text generated by each model. These tasks were structured to test the LLMs on the same set of tasks with increasing difficulty factors. Each task involved generating text in response to prompts of varying complexity, covering a range of linguistic styles, topics, and contextual constraints. To ensure the reliability and validity of the experimental design, we carefully curated the prompts and task conditions to control for potential confounding variables. Moreover, we randomised the order of tasks and counterbalanced the assignment of LLMs to minimise order effects and bias in the evaluation process.
   The experimental tasks were designed to simulate real-world scenarios where the readability of generated text is crucial, such as educational materials, news articles, and user interface content. By systematically varying the complexity of the tasks, we aimed to assess the robustness and adaptability of each LLM in producing readable text across different contexts and linguistic challenges.

Through this methodological approach, we aimed to provide a comprehensive evaluation of the readability performance of LLMs and identify factors influencing their efficacy in generating human-readable text. The results of this

investigation will contribute valuable insights into the strengths and limitations of LLMs in addressing the readability challenges inherent in automated text generation tasks.

The study has been performed considering:

- The specific type of LLM being studied (e.g., Transformer-based models);
- The methodology used to assess readability (e.g., human evaluation, automated metrics);
- The aim of highlighting the main differences between the models, evaluated on the same task.

Ultimately, the best text, automatically generated by the LLMs, should be concise, informative, and accurately reflect the content of the users' requests. The latter are represented by the different tasks we submitted to the pool of LLM. In particular, the pair (type of text, number of words) is a key characteristic to discriminate the readability of the generated texts.

4. Experimental Design and Results Discussion

In this section are provided: i) the detailed definition of the tasks required to be performed by each LLMs, along with and the supporting hardware architectures; and ii) a preliminary discussion about the results of this experimental campaign.

4.1. Task Definition

The experimental approach involved conducting a series of tests where each LLM created three different writings, precisely technical, science fiction, and journalism content. Each text required three variants of different lengths, specifically 1000, 500, and 250 words. This approach facilitated a thorough assessment of the LLMs' capacity to produce logical and contextually appropriate content across various genres and lengths, primarily evaluated in terms of readability. The LLMs adopted to perform this study are: Microsoft phy 3.5B, Llama3 8B, Mistral 8x7B, Llama3 70B, OpenAI ChatGPT 3.5.

The selection of these models was based on their different sizes, typologies of contents field, and training aims, enabling the evaluation of a wide variety of language generation skills. To guarantee the consistency and effectiveness of the experimental campaign, the Python library Langchain[1] was employed. Langchain offers a consolidated interface for interacting with different LLMs, facilitating easy integration and execution of the models.

The numerous LLM systems were implemented on diverse hardware architectures, including local and cloud-based infrastructure, dependent on their level of complexity. The Microsoft phy 3.5B and Llama 3 8B models were run on an Apple MacBook Pro M1 computer with 16 GB of unified RAM using the Ollama software. This configuration facilitated efficient processing and rapid iteration during the experimental phase. On the other hand, the bigger models were utilised on the Groq cloud platform, which offered expandable computational resources and exceptional processing capabilities. The OpenAI ChatGPT 3.5 model was invoked using its API, which enabled effortless integration with the experimental framework. Table 1 shows the prompts used to generate the contents. After this procedure, we have obtained a total of 54 distinct texts to examine.

During the second phase of this experimental study, four readability indexes were utilised (as previously illustrated in Section 3) to acquire results for each of the texts that had been prepared as part of the first investigation. This approach made it possible to conduct a comprehensive evaluation of the LLMs' ability to write content that is logical and contextually suitable over a wide range of genres and lengths, with the primary focus being on the readability of the content under consideration.

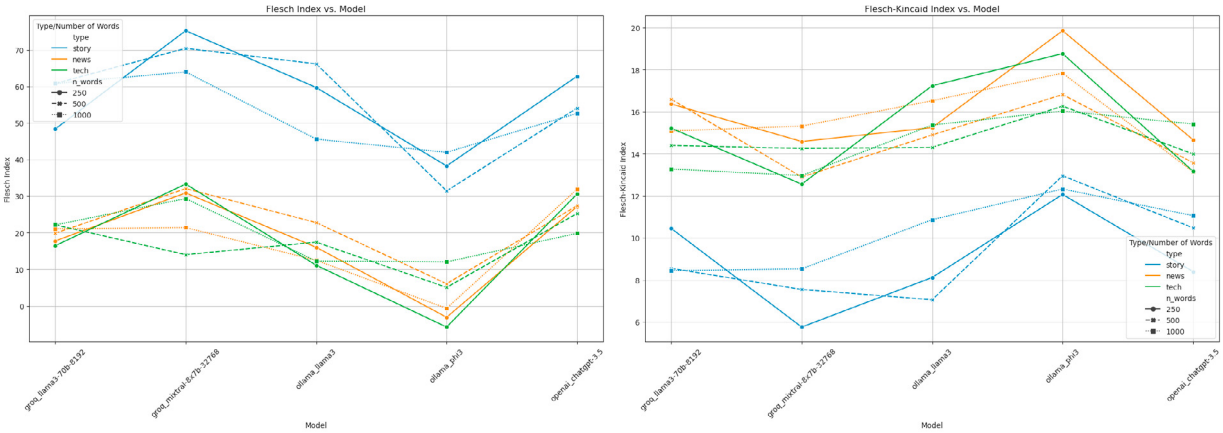
4.2. Results Inspection and Discussion

To collect observations and evaluate the results of the experiments, it is possible to analyse the graphs collected from Figure 1. According to the FRE Index, referring to Figure 1a, for all generative models, the complexity of texts

[1] https://python.langchain.com/v0.2/docs/introduction/

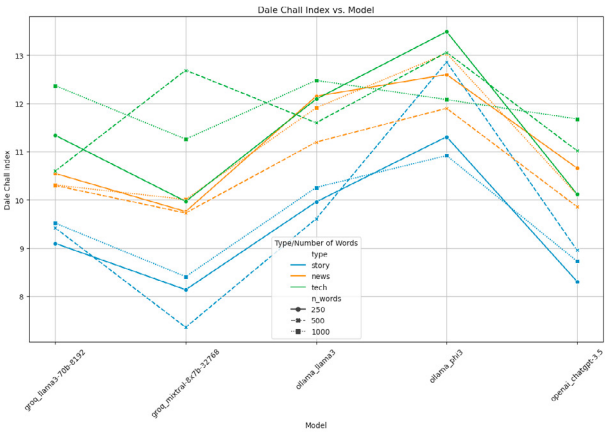
Prompt	Max Words	Type
Act as an experienced writer to write a science fiction short story	1000	Story
Act as an experienced writer to write a science fiction short story	500	Story
Act as an experienced writer to write a science fiction short story	250	Story
Act as a journalist to write a news article on the use of blockchain to mine new coins	1000	News
Act as a journalist to write a news article on the use of blockchain to mine new coins	500	News
Act as a journalist to write a news article on the use of blockchain to mine new coins	250	News
Act as an experienced technical writer to write an essay on the pros and cons of using blockchain to protect privacy	1000	Tech
Act as an experienced technical writer to write an essay on the pros and cons of using blockchain to protect privacy	500	Tech
Act as an experienced technical writer to write an essay on the pros and cons of using blockchain to protect privacy	250	Tech

Table 1: Prompts used in LLMs

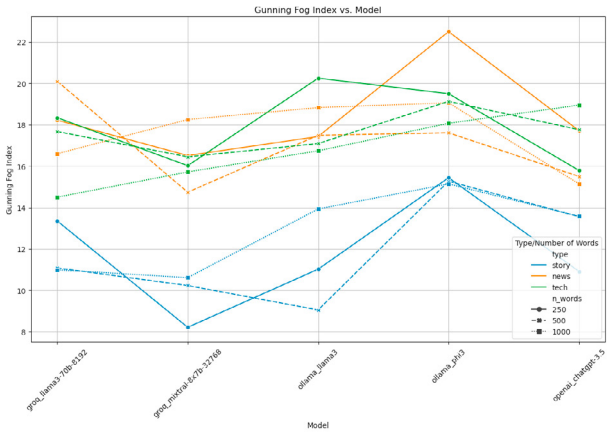


(a) FRE Index

(b) FKGL Index



(c) Dale-Chall Index



(d) GF Index

Fig. 1: Formulas vs Models

increases when, given the pair (type of text, number of words), type of text is fixed on “story” and the number of

model	n_words	type	flesch_kincaid	flesch	gunning_fog	dale_chall
ollama_phi3	1000	story	12.33	41.99	15.14	10.92
ollama_phi3	500	story	12.96	31.44	15.27	12.86
ollama_phi3	250	story	12.07	38.26	15.44	11.31
ollama_phi3	1000	news	17.84	-0.67	19.06	13.04
ollama_phi3	500	news	16.82	6.06	17.61	11.9
ollama_phi3	250	news	19.86	-3.1	22.5	12.6
ollama_phi3	1000	tech	16.04	12.03	18.07	12.08
ollama_phi3	500	tech	16.27	5.07	19.13	13.06
ollama_phi3	250	tech	18.77	-5.78	19.5	13.49
ollama_llama3	1000	story	10.87	45.6	13.93	10.26
ollama_llama3	500	story	7.06	66.18	9.05	9.61
ollama_llama3	250	story	8.12	59.7	11.02	9.96
ollama_llama3	1000	news	16.53	12.48	18.83	11.91
ollama_llama3	500	news	14.9	22.79	17.49	11.2
ollama_llama3	250	news	15.25	16.04	17.43	12.15
ollama_llama3	1000	tech	15.37	12.26	16.74	12.48
ollama_llama3	500	tech	14.31	17.43	17.09	11.6
ollama_llama3	250	tech	17.24	11.12	20.25	12.09
groq_llama3-70b-8192	1000	story	8.44	60.78	10.99	9.52
groq_llama3-70b-8192	500	story	8.55	60.79	11.08	9.42
groq_llama3-70b-8192	250	story	10.45	48.36	13.35	9.1
groq_llama3-70b-8192	1000	news	15.09	21.1	16.6	10.31
groq_llama3-70b-8192	500	news	16.6	19.88	20.1	10.3
groq_llama3-70b-8192	250	news	16.37	17.76	18.22	10.55
groq_llama3-70b-8192	1000	tech	13.28	22.14	14.5	12.37
groq_llama3-70b-8192	500	tech	14.4	22.16	17.67	10.6
groq_llama3-70b-8192	250	tech	15.22	16.46	18.34	11.34
groq_mixtral-8x7b-32768	1000	story	8.53	63.96	10.61	8.41
groq_mixtral-8x7b-32768	500	story	7.55	70.41	10.23	7.36
groq_mixtral-8x7b-32768	250	story	5.76	75.23	8.21	8.14
groq_mixtral-8x7b-32768	1000	news	15.32	21.42	18.25	10.01
groq_mixtral-8x7b-32768	500	news	12.9	32.06	14.74	9.73
groq_mixtral-8x7b-32768	250	news	14.59	30.89	16.51	9.76
groq_mixtral-8x7b-32768	1000	tech	12.97	29.34	15.72	11.26
groq_mixtral-8x7b-32768	500	tech	14.26	14.01	16.45	12.69
groq_mixtral-8x7b-32768	250	tech	12.56	33.32	16.02	9.97
openai_chatgpt-3.5	1000	story	11.06	52.62	13.57	8.73
openai_chatgpt-3.5	500	story	10.48	54.06	13.56	8.96
openai_chatgpt-3.5	250	story	8.4	62.78	10.91	8.31
openai_chatgpt-3.5	1000	news	13.15	31.87	15.14	10.11
openai_chatgpt-3.5	500	news	13.58	27.41	15.5	9.86
openai_chatgpt-3.5	250	news	14.67	27.15	17.71	10.66
openai_chatgpt-3.5	1000	tech	15.42	19.84	18.95	11.68
openai_chatgpt-3.5	500	tech	14.0	25.26	17.76	11.02
openai_chatgpt-3.5	250	tech	13.17	30.59	15.79	10.12

Table 2: Experimental Results

words is left variable. Moreover, the most difficult story to read is the one generated by groq_mixtral-8x7b-32768 with a constraint of 250 words.

On the other hand, analysing the graph for the FKGL Index, in Figure 1b, "news" is the type of text with higher readability than the other two. This time, the ollama_phi3 model generates the more complex text with a constraint of 250 words. The same result can be appreciated in the graph, Figure 1d, obtained from analysing the GF index values.

For the Dale-Chall Index, shown in Figure 1c, the most complex type of text is the tech one, but again the peak is reached with a "250 words" request performed by the ollama_phi3 model.

Summarising, analyzing the graphs it can be observed that the most influential constraint affecting the complexity of the text (in terms of readability) is the number of words when it is set to 250 words. In fact, when we consider text types like "news" and "tech", this constraint becomes very stringent, even for a human expert: a smaller number of words requires compacting concepts, thus making argumentation more difficult. On the contrary, this constraint makes the model groq_mixtral-8x7b-32768 working better with the "story" type.

What can be observed is that the type "story", that does not pertain to technical topics or informative documents, turns out to have more flexible content to generate by adapting better to the constraint of words. In fact, a "story" increases in complexity the higher the number of words it contains: this aspect is evidenced in the graphs, when looking at the "1000 words" parameter.

5. Conclusions and Future Work

Training Large Language Models (LLMs) enables the performance of various tasks, including writing different types of creative text formats, translating languages, and even writing basic computer code. LLMs are continually evolving and hold significant promise for the future of AI. Assessing the quality of the generated texts in terms of content reliability, lexical variety, and text coherence is crucial.

This work provides a case study focused on analyzing the content generation capabilities of several LLMs, attempting to identify their strengths, shortcomings, and potential applications in different domains.

A set of experiments have been performed for providing preliminary observations useful to evaluate potential behavioral differences among various pre-trained LLMs. Specifically, a key goal is to highlight how differently these LLMs perform on a constrained text generation task. The constraints involve summarizing content without increasing text complexity. Variations in text complexity can be evaluated using readability indexes, as previously introduced. To achieve the goal, as the evaluation metrics to perform this analysis, the readability of texts was adopted. Several pre-trained LLMs were employed to analyze a pool of English text generation tasks. The readability of the artificial texts was assessed using various indices, highlighting the influence of word count constraints and the story on text complexity.

So, a significant insight provided by this study is found in the opportunity to compare the behavior of selected LLMs and to assess how these LLMs select content to retain and how the readability of a text varies when constraints are imposed on the text's characteristics. In the experiment, the number of words allowed in the text is gradually reduced. It is expected that as the number of available words decreases, readability might also decrease. This hypothesis is preliminarily supported in two out of three cases considered as for the experiments. Depending on the "type" of content that is asked to be generated, different trends of the readability parameter can be observed. This aspect need to be further investigated but it can be reasonable explained by observing that reducing the number of words available to express or describe a concept, might lead to the use of lexically or semantically denser terms, such as technical jargon where individual terms encapsulate well-defined concepts.

Finally, it is important to note that relying solely on the readability metrics can be limiting, as they may not capture all nuances of readability, such as contextual appropriateness and reader engagement. These challenges will need to be addressed in future plans.

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